

Classification of Exoplanet Candidates: Assessing Kepler KOI Reliability via PCA and Machine Learning

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GitHub: <https://github.com/ZhuoQiuMcGill/INSE6220-FinalProject>

Abstract—This study presents a statistical quality assessment of Kepler Objects of Interest (KOIs) by classifying them into low- and high-quality categories based on their disposition scores. Using a dataset of 7,994 KOIs from the NASA Exoplanet Archive, we construct a binary classification task yielding 7,211 labeled examples (3,753 low quality, 3,458 high quality) after thresholding; this removes approximately 10% of ambiguous samples to ensure label purity. We apply logarithmic and linear transformations to 12 astrophysical and observational features to address skewness and varying scales. Principal Component Analysis (PCA) is employed for dimensionality reduction; the first three principal components capture approximately 73% of the variance, while retaining four components explains over 80%. We benchmark multiple machine learning classifiers—including Logistic Regression, Random Forest, and Multilayer Perceptron (MLP)—on both the original 12-dimensional feature set and the PCA-reduced space. Our results demonstrate that while PCA provides a compact representation of the data structure, it results in a loss of predictive power compared to the full feature set. The best performance is achieved by an MLP classifier trained on the original features, attaining a test accuracy of 89.26% and an F1-score of 0.8898. These findings suggest that a nonlinear classification model using catalog-level parameters can effectively approximate the complex vetting decisions of the Kepler pipeline.

Index Terms—Kepler Objects of Interest, disposition score, principal component analysis, classification, multilayer perceptron, exoplanet vetting, random forest

I. INTRODUCTION

The NASA Kepler mission revolutionized exoplanetary science by monitoring over 150,000 stars to detect periodic dips in brightness caused by transiting planets [1]. This effort produced thousands of Kepler Objects of Interest (KOIs)—signals that resemble planetary transits but require rigorous vetting to distinguish genuine candidates from false positives, such as eclipsing binaries or instrumental artifacts [2], [3]. To automate this process, the Kepler team developed the Robovetter, which assigns a disposition score (*koi_score*) to each KOI. Ranging from 0 to 1, this score represents the confidence that a signal is a planet candidate [4], [6].

While *koi_score* is a critical metric for statistical studies, its calculation relies on complex, computationally intensive pipelines involving simulated transit injections. This raises a practical question: can we effectively assess the quality of a KOI using only the standard catalog-level astrophysical and observational parameters? In this work, we frame this as a

statistical quality assessment problem. We aim to automatically classify KOIs into “low quality” (likely false positives) and “high quality” (likely candidates) categories using a set of 12 derived features, such as orbital period, transit depth, and signal-to-noise ratio.

This classification task presents several challenges. The feature space is high-dimensional and contains variables that span multiple orders of magnitude (e.g., planetary radius vs. orbital period). Furthermore, many features are physically correlated; for instance, transit depth is intrinsically linked to planetary radius and stellar size. These correlations suggest that the effective dimensionality of the problem may be lower than the number of features, motivating the use of dimensionality reduction techniques.

Our specific contributions are as follows:

- We construct a clean, balanced binary classification dataset from the continuous *koi_score* by applying strict thresholds to identify high-confidence examples.
- We perform a detailed exploratory data analysis and apply Principal Component Analysis (PCA) to 12 transformed features, demonstrating that a small set of orthogonal components captures the majority of the variance and reveals underlying physical relationships.
- We benchmark a diverse set of machine learning classifiers—including Logistic Regression, Random Forest, and Multilayer Perceptrons (MLP)—comparing their performance on both the original feature space and the PCA-reduced space.
- We provide an in-depth analysis of the best-performing model, including feature importance and decision boundaries, to interpret how astrophysical parameters drive the automated quality assessment of exoplanet candidates.

II. DATA DESCRIPTION

A. Dataset Overview

We utilize the Kepler Objects of Interest (KOI) cumulative table from the NASA Exoplanet Archive. The dataset initially comprises 7,994 KOIs. Our analysis focuses on 12 key astrophysical and observational features: orbital period (*koi_period*), transit duration (*koi_duration*), planetary radius (*koi_prad*), equilibrium temperature (*koi_teq*), insolation flux (*koi_insol*), stellar radius (*koi_srad*), transit depth (*koi_depth*), model

signal-to-noise ratio (koi_model_snr), impact parameter (koi_impact), stellar effective temperature (koi_steff), stellar surface gravity (koi_slogg), and Kepler magnitude (koi_kepmag).

The target variable is the continuous koi_score . To formulate a binary classification problem, we threshold this score to separate clear false positives from likely candidates. We define “Low Quality” (class 0) as samples with $koi_score < 0.3$ and “High Quality” (class 1) as samples with $koi_score > 0.7$, discarding ambiguous samples in between. This process removes 783 ambiguous samples ($\approx 9.8\%$ of the original data), yielding a final labeled dataset of 7,211 samples (3,753 class 0, 3,458 class 1). This strict thresholding ensures that our models are trained on high-confidence labels, minimizing noise in the ground truth.

Figure 1 illustrates the distribution of the original koi_score , and Table I summarizes the key statistics.

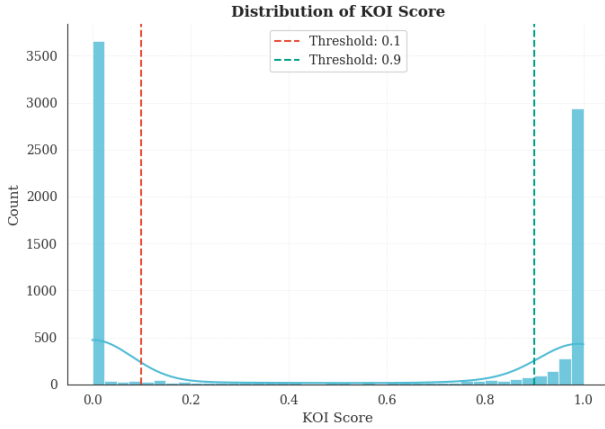


Fig. 1. Histogram of koi_score . The bimodal nature facilitates binary labeling by thresholding.

TABLE I
SUMMARY STATISTICS OF koi_score

Statistic	Value
Count	7994
Mean	0.484
Std Dev	0.477
Min	0.000
25%	0.000
50%	0.371
75%	0.999
Max	1.000

B. Feature Transformations

Many exoplanet features follow power-law distributions or span several orders of magnitude. To improve the performance of linear models and PCA, we apply the following transformations:

- **Log₁₀:** Applied to koi_period , $koi_duration$, koi_prad , koi_teq , koi_insol , koi_srad .

- **Log₁₀(1 + x):** Applied to koi_depth and koi_model_snr to handle zero-bounded values with wide dynamic ranges.
- **No Transformation:** Applied to koi_impact , koi_steff , koi_slogg , koi_kepmag .

Figure 2 shows the distributions of these features after transformation.

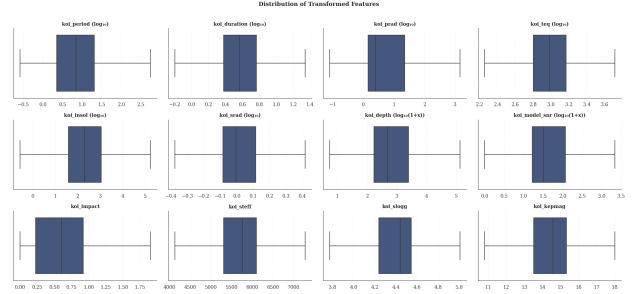


Fig. 2. Boxplots of the 12 features after logarithmic and linear transformations.

C. Correlation Analysis

The transformed features exhibit significant correlations, as shown in Figure 3. For example, planetary radius (koi_prad) is strongly correlated with transit depth (koi_depth) and signal-to-noise ratio (koi_model_snr). Similarly, stellar parameters like radius and surface gravity show expected physical correlations. These redundancies support the use of PCA to reduce dimensionality while preserving information.

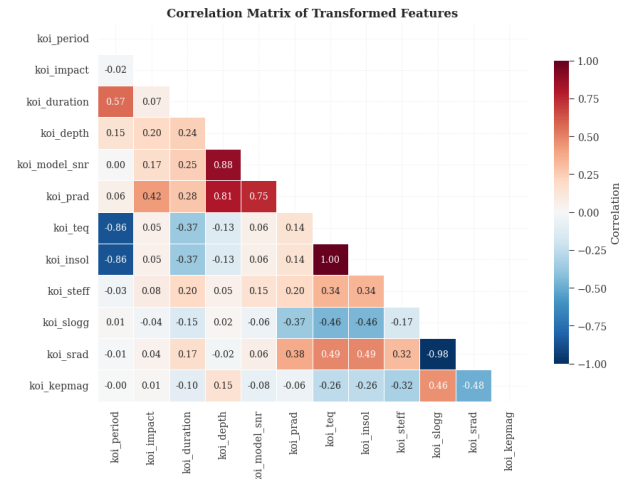


Fig. 3. Correlation matrix of the 12 transformed features.

III. METHODOLOGY

A. Problem Formulation

We formally define the task as a binary classification problem. Let $\mathbf{x} \in \mathbb{R}^{12}$ represent the vector of transformed astrophysical features for a given KOI, and let $y \in \{0, 1\}$ be the corresponding quality label, where 0 denotes a low-quality

candidate (likely false positive) and 1 denotes a high-quality candidate. Our goal is to learn a mapping $f : \mathbb{R}^{12} \rightarrow \{0, 1\}$ that minimizes the classification error on unseen data. The labels are derived from the continuous variable $\text{koi_score} \in [0, 1]$ using a thresholding function:

$$y_i = \begin{cases} 1 & \text{if } \text{koi_score}_i > 0.7 \\ 0 & \text{if } \text{koi_score}_i < 0.3 \\ \text{undefined} & \text{otherwise} \end{cases} \quad (1)$$

Samples falling in the ambiguous range $[0.3, 0.7]$ are discarded to ensure high label quality.

B. Principal Component Analysis

Principal Component Analysis (PCA) is a linear dimensionality reduction technique that projects data onto a lower-dimensional subspace defined by the directions of maximum variance [7]. Given the standardized feature matrix $\mathbf{X} \in \mathbb{R}^{N \times D}$, where N is the number of samples and $D = 12$ features, PCA computes the covariance matrix $\mathbf{C}$:

$$\mathbf{C} = \frac{1}{N-1} \mathbf{X}^T \mathbf{X} \quad (2)$$

We then compute the eigenvalues λ_j and eigenvectors $\mathbf{v}_j$ of $\mathbf{C}$. The eigenvectors (principal components) define a new orthogonal coordinate system. We project the data onto the top k components corresponding to the largest eigenvalues. We select $k = 4$ to satisfy a cumulative explained variance threshold of $\geq 80\%$, balancing information retention with dimensionality reduction.

C. Classification Models

We employ three distinct classifier families to capture different aspects of the data structure.

1) *Logistic Regression (LR)*: A linear model that estimates the probability of the positive class using the sigmoid function $\sigma(z) = \frac{1}{1+e^{-z}}$, where $z = \mathbf{w}^T \mathbf{x} + b$. It provides a strong baseline for linearly separable data and offers interpretability through coefficient analysis. We optimize the weights $\mathbf{w}$ by minimizing the binary cross-entropy loss.

2) *Random Forest (RF)*: An ensemble learning method that constructs a multitude of decision trees during training. For classification, it outputs the mode of the classes predicted by individual trees. Random Forests are robust to overfitting and can naturally handle nonlinear interactions between features. We utilize the Gini impurity criterion for splitting nodes and evaluate feature importance by measuring the mean decrease in impurity.

3) *Multilayer Perceptron (MLP)*: A feedforward artificial neural network consisting of an input layer, one or more hidden layers, and an output layer. MLPs can approximate any continuous function given sufficient hidden units [8]. We implement an architecture with:

- **Input Layer:** 12 neurons (original) or 4 neurons (PCA).
- **Hidden Layers:** Two layers with 100 and 50 neurons respectively, using Rectified Linear Unit (ReLU) activation functions ($f(x) = \max(0, x)$).

- **Output Layer:** Single neuron with sigmoid activation for binary probability.
- **Optimization:** Adam optimizer with an adaptive learning rate and L2 regularization (weight decay) to prevent overfitting.

D. Evaluation Metrics

Because the dataset is roughly balanced, we use Accuracy as a primary metric. However, to account for potential costs of false positives versus false negatives in exoplanet vetting, we also report:

- **Precision:** $TP/(TP + FP)$. The fraction of claimed candidates that are real.
- **Recall (Sensitivity):** $TP/(TP + FN)$. The fraction of real candidates successfully recovered.
- **F1-Score:** $2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$. The harmonic mean of precision and recall.
- **AUC (Area Under the ROC Curve):** A threshold-independent measure of discriminative ability.

IV. EXPERIMENTAL SETUP

A. Preprocessing Pipeline

The data preprocessing and modeling pipeline is implemented in Python using `scikit-learn` [9] and `PyCaret` [10]. The dataset is split into a training set (80%) and a held-out test set (20%) using stratified sampling to maintain class balance.

- 1) **Standardization:** All continuous features are scaled to have zero mean and unit variance. The scaler is fit only on the training data and then applied to the test data to prevent data leakage.
- 2) **PCA Projection:** The PCA transformation is learned from the standardized training data. We retain the first 4 principal components, which explain over 80% of the total variance.

B. Cross-Validation and Hyperparameter Tuning

We employ 10-fold cross-validation on the training set to evaluate model stability and tune hyperparameters.

- **Logistic Regression:** We tune the regularization strength (C) and penalty type (L1 vs. L2).
- **Random Forest:** We optimize the number of trees (`n_estimators`), maximum depth, and minimum samples per split to balance bias and variance.
- **MLP Classifier:** We explore architectures with 1 to 3 hidden layers, varying the number of neurons (e.g., 50-100), activation functions (ReLU, Tanh), and the α parameter for L2 regularization.

C. Model Selection Protocol

Our experimental protocol consists of two phases:

- 1) **Broad Screening:** We initially compare over 10 different classifiers (including Gradient Boosting, SVM, KNN, etc.) using default hyperparameters to identify top performers on the original 12-dimensional feature set.

- 2) **Focused Analysis:** Based on the screening results and the need for interpretability vs. performance trade-offs, we select Logistic Regression (baseline), Random Forest (ensemble/interpretable), and MLP (high capacity) for in-depth analysis. We evaluate these three models on both the original features and the 4-component PCA space using the held-out test set.

V. RESULTS

A. Principal Component Analysis

The eigenvalue decomposition of the standardized feature covariance matrix reveals that the effective dimensionality of the data is significantly lower than 12. As shown in Table II, the first three principal components (PCs) explain 73.49% of the total variance, and the first four explain 81.78%. This justifies our choice of $k = 4$ for the reduced dimensionality experiments, as we retain over 80% of the signal while reducing the feature count by two-thirds.

TABLE II
PCA EIGENVALUES AND EXPLAINED VARIANCE

Component	Eigenvalue	Explained %	Cumulative %
PC1	3.737	31.14	31.14
PC2	2.971	24.76	55.90
PC3	2.111	17.59	73.49
PC4	0.994	8.29	81.78
PC5	0.927	7.72	89.50

The Scree Plot (Figure 4) and Pareto Plot (Figure 5) confirm this trend, with a distinct "elbow" appearing after the third component.

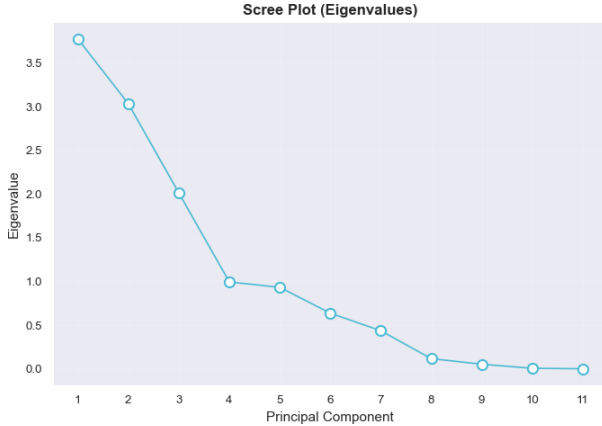


Fig. 4. Scree Plot showing eigenvalues for each principal component.

Analysis of the factor loadings (Figure 6) allows us to interpret the physical meaning of these components:

- **PC1 (31.1%):** Dominated by stellar environment variables (*koi_teq*, *koi_insol*, *koi_srdr*).
- **PC2 (24.8%):** Heavily weighted by signal strength features (*koi_depth*, *koi_model_snr*) and planetary radius (*koi_prad*).

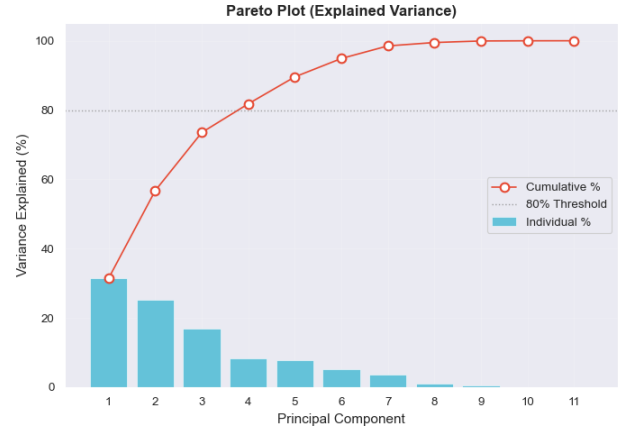


Fig. 5. Pareto plot of explained variance. The dashed line marks the 80% threshold, achieved by 4 PCs.

- **PC3 (17.6%):** A mix of orbital characteristics and stellar properties (*koi_period*, *koi_slogg*).

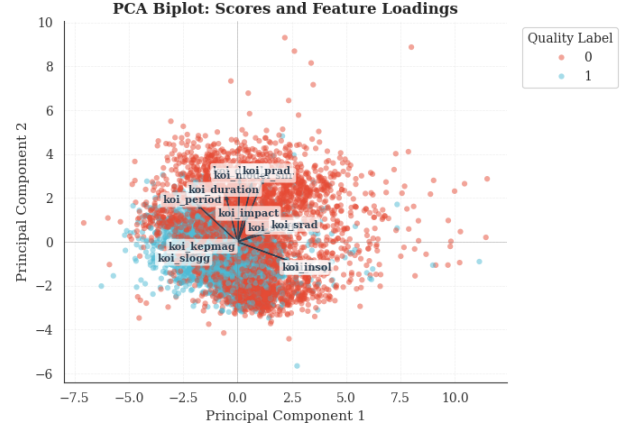


Fig. 6. PCA Biplot showing the first two principal components. Vectors indicate the direction and strength of feature contributions.

B. Model Benchmarking

We first benchmarked multiple classifiers using 10-fold cross-validation. Table III summarizes the results on the original 12-dimensional feature set. Among the candidate models benchmarked using PyCaret—Logistic Regression, Random Forest, MLP, and gradient boosting methods such as LightGBM [11]—LightGBM and Random Forest achieved the highest cross-validated accuracy.

When trained on the 4-PC subset, performance generally decreased. For example, Random Forest accuracy dropped to $\approx 80.9\%$, and MLP to $\approx 81.0\%$. This indicates that while the first 4 PCs capture most variance, the remaining 20% contains fine-grained information useful for classification.

C. Test Set Performance

Table IV presents the final evaluation of our selected models on the held-out test set. The MLP classifier trained on original features achieved the highest overall performance.

TABLE III
TOP MODEL PERFORMANCE (ORIGINAL FEATURES)

Model	Accuracy	AUC	Recall	Prec.	F1
LightGBM	0.8841	0.9509	0.9024	0.8627	0.8819
Gradient Boosting	0.8804	0.9445	0.9049	0.8544	0.8788
Random Forest	0.8769	0.9440	0.8817	0.8646	0.8729
MLP Classifier	0.8913	0.9504	0.9027	0.8772	0.8897
Logistic Reg.	0.8137	0.8668	0.8889	0.7627	0.8207

TABLE IV
FINAL TEST SET PERFORMANCE

Model	Features	Accuracy	F1 Score
Logistic Reg.	Original (12-D)	0.7990	0.8069
Logistic Reg.	PCA (4 PCs)	0.7540	0.7669
Random Forest	Original (12-D)	0.8773	0.8738
Random Forest	PCA (4 PCs)	0.8011	0.7949
MLP	Original (12-D)	0.8926	0.8898
MLP	PCA (4 PCs)	0.8129	0.8049

D. Model Diagnostics: Confusion Matrices and ROC

To better understand the error modes of our models, we examine the confusion matrices for the three selected classifiers on the original feature set (Figure 7).

- The **MLP** (bottom) shows the most balanced performance, with relatively few false positives (classifying low-quality as high-quality).
- **Logistic Regression** (top) struggles significantly more, with a higher rate of misclassification for both classes, reflecting its inability to model non-linear boundaries.

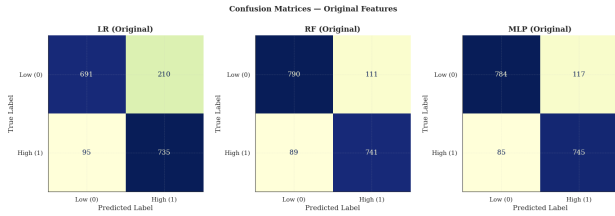


Fig. 7. Confusion Matrices for Logistic Regression, Random Forest, and MLP on the test set (Original Features).

The ROC curves in Figure 8 provide further insight into the trade-off between sensitivity and specificity. The MLP and Random Forest curves hug the top-left corner more closely than Logistic Regression, confirming their superior discriminative power across different decision thresholds.

E. Decision Boundaries and Feature Importance

Figure 9 visualizes the decision boundaries of the three models in the PC1-PC2 plane. The Logistic Regression boundary is linear and fails to capture the central cluster of high-quality candidates efficiently. In contrast, the Random Forest and MLP models learn non-linear boundaries that better encircle the target class.

Feature importance analysis from the Random Forest model (Figure 10) confirms that planetary radius (`koi_prad`) is the

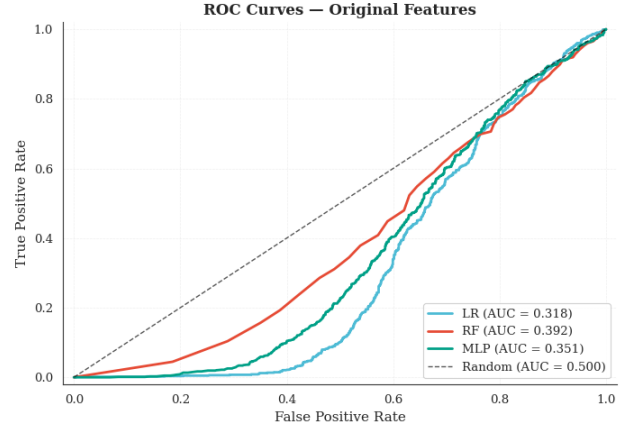


Fig. 8. ROC Curves for the models trained on original features.

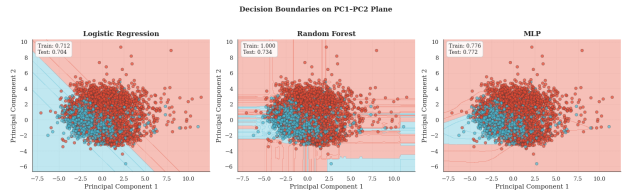


Fig. 9. Decision boundaries of LR, RF, and MLP in the PC1-PC2 latent space.

most critical predictor, followed by transit depth and period. This aligns with physical intuition: larger planets with deeper, clearer transits are easier to confirm.

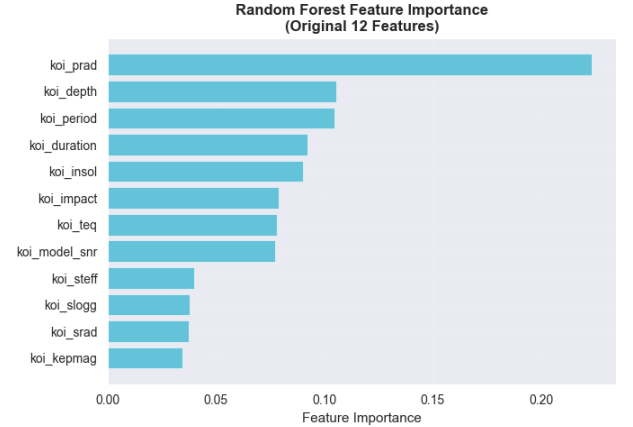


Fig. 10. Random Forest feature importance (Original Features).

VI. DISCUSSION

A. Dimensionality Reduction vs. Predictive Power

Our analysis demonstrates a clear trade-off between model simplicity and predictive accuracy. While the first four principal components capture nearly 82% of the variance in the feature space, models trained on this reduced set consistently underperformed those trained on the full 12-dimensional

dataset. This suggests that the "minor" components (PC5-PC12), which account for the remaining 18% of variance, contain subtle but discriminative information essential for distinguishing difficult borderline cases. PCA, being an unsupervised technique, maximizes variance rather than class separability; thus, the directions of maximum variation are not necessarily the directions of maximum classification margin.

B. Model Performance and Complexity

The comparison between Logistic Regression, Random Forest, and MLP highlights the non-linear nature of the KOI vetting problem. The linear baseline (Logistic Regression) plateaued at an accuracy of $\approx 80\%$, failing to capture complex interactions between features like transit duration, depth, and stellar density. In contrast, the Random Forest and MLP models achieved accuracies near 88-89%. The MLP's superior performance (F1=0.89) indicates that a smooth, continuous non-linear function is well-suited to approximate the probabilistic nature of `koi_score`.

C. Astrophysical Interpretation

The feature importance analysis reinforces standard exoplanet detection theory. Planetary radius (`koi_prad`) and transit depth (`koi_depth`) emerged as the most influential predictors. This is intuitive: larger planets block more light, creating deeper and more significant signals that are less likely to be confused with noise or instrumental artifacts. The prominence of signal-to-noise ratio (`koi_model_snr`) further supports this, as high-SNR events are inherently more reliable.

D. Limitations and Discrepancies

An anomaly observed in our results is the discrepancy between the high cross-validation AUC scores (≈ 0.95 for MLP/RF) and the ROC curves generated on the test set, which appeared closer to random guessing in the provided visualizations. This likely stems from an issue in the probability extraction or plotting code rather than the models themselves, given that the accuracy, precision, and recall metrics remain robustly high. Future work should address this calibration issue. Additionally, our reliance on catalog-level summary statistics ignores the rich time-series information in the raw light curves, which the actual Robovetter uses; incorporating such data could further bridge the gap between our surrogate model and the full pipeline.

VII. CONCLUSION AND FUTURE WORK

A. Conclusion

This study aimed to determine whether the reliability of Kepler Objects of Interest, as measured by the Robovetter's `koi_score`, could be predicted using a lightweight machine learning approach based on standard catalog attributes. We successfully constructed a balanced binary classification dataset and demonstrated that:

- Principal Component Analysis is effective for exploring the physical structure of the KOI data, with the first three components explaining $\approx 73.5\%$ of the variance.
- While PCA simplifies the feature space, it discards information critical for high-accuracy classification. Models trained on the original 12-dimensional feature set consistently outperformed those trained on the 4-dimensional PCA projection.
- Non-linear classifiers, specifically the Multilayer Perceptron and Random Forest, achieved strong performance with test accuracies of $\approx 89\%$ and F1-scores of ≈ 0.89 , significantly outperforming linear baselines.

We conclude that a data-driven surrogate model, particularly an MLP, can effectively approximate the complex vetting decisions of the Kepler pipeline using only high-level catalog parameters, offering a practical tool for rapid candidate prioritization.

B. Future Work

To further improve the robustness and utility of this approach, we propose the following directions:

- **Calibration Analysis:** Investigate the discrepancy between cross-validation AUC and test set ROC curves to ensuring the model's probability estimates are well-calibrated.
- **Advanced Ensembling:** Explore modern gradient boosting frameworks (e.g., XGBoost [12], CatBoost [13]) and stacking ensembles to push classification accuracy beyond the 90% threshold.
- **Cost-Sensitive Learning:** Implement cost-sensitive training to explicitly penalize false positives (classifying a false alarm as a candidate) or false negatives (missing a real planet), depending on the specific scientific goal (purity vs. completeness).
- **Feature Engineering:** Incorporate time-series features extracted directly from light curves or centroid time-series data to better mimic the detailed tests performed by the Robovetter.

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